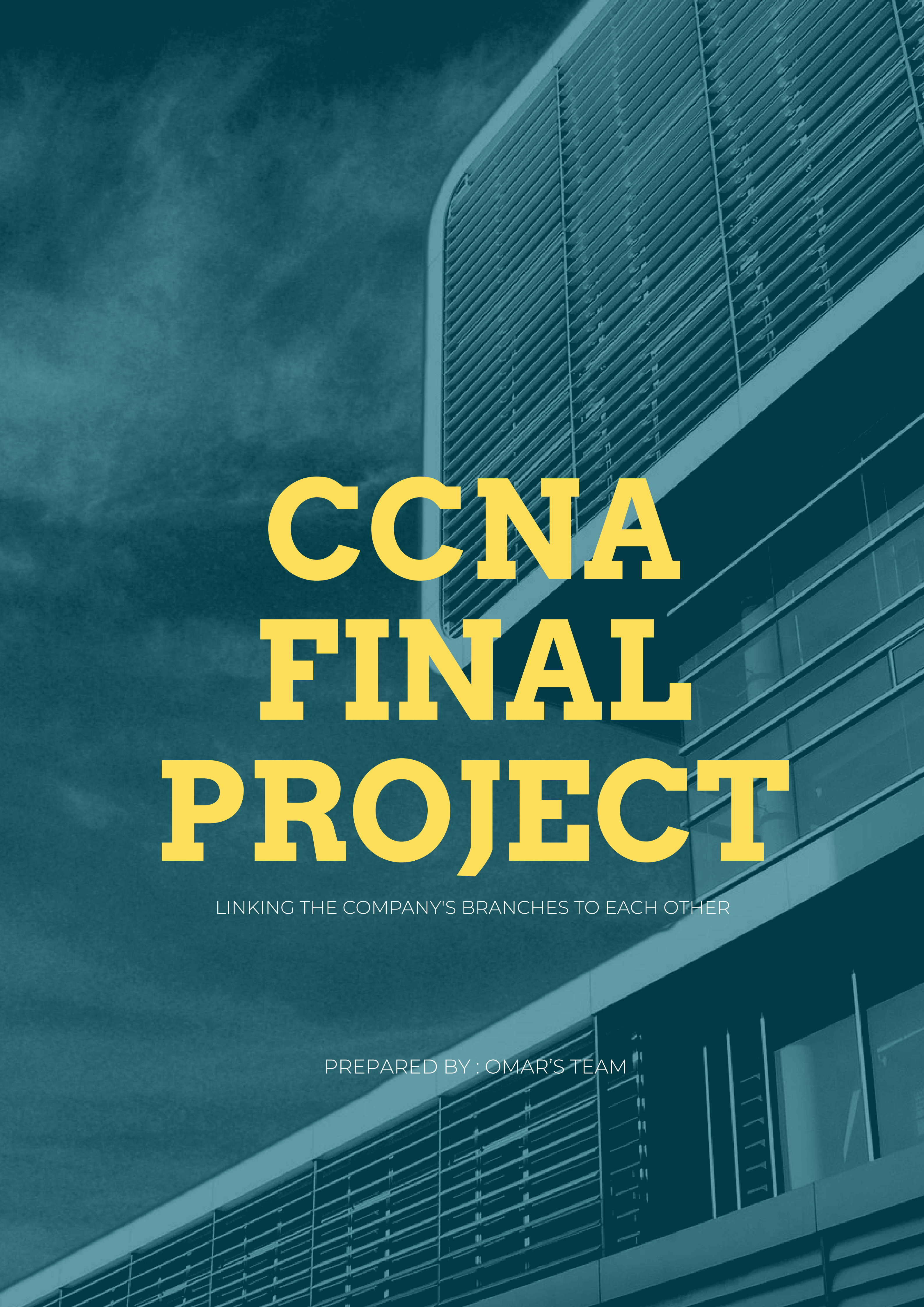


CCNA FINAL PROJECT

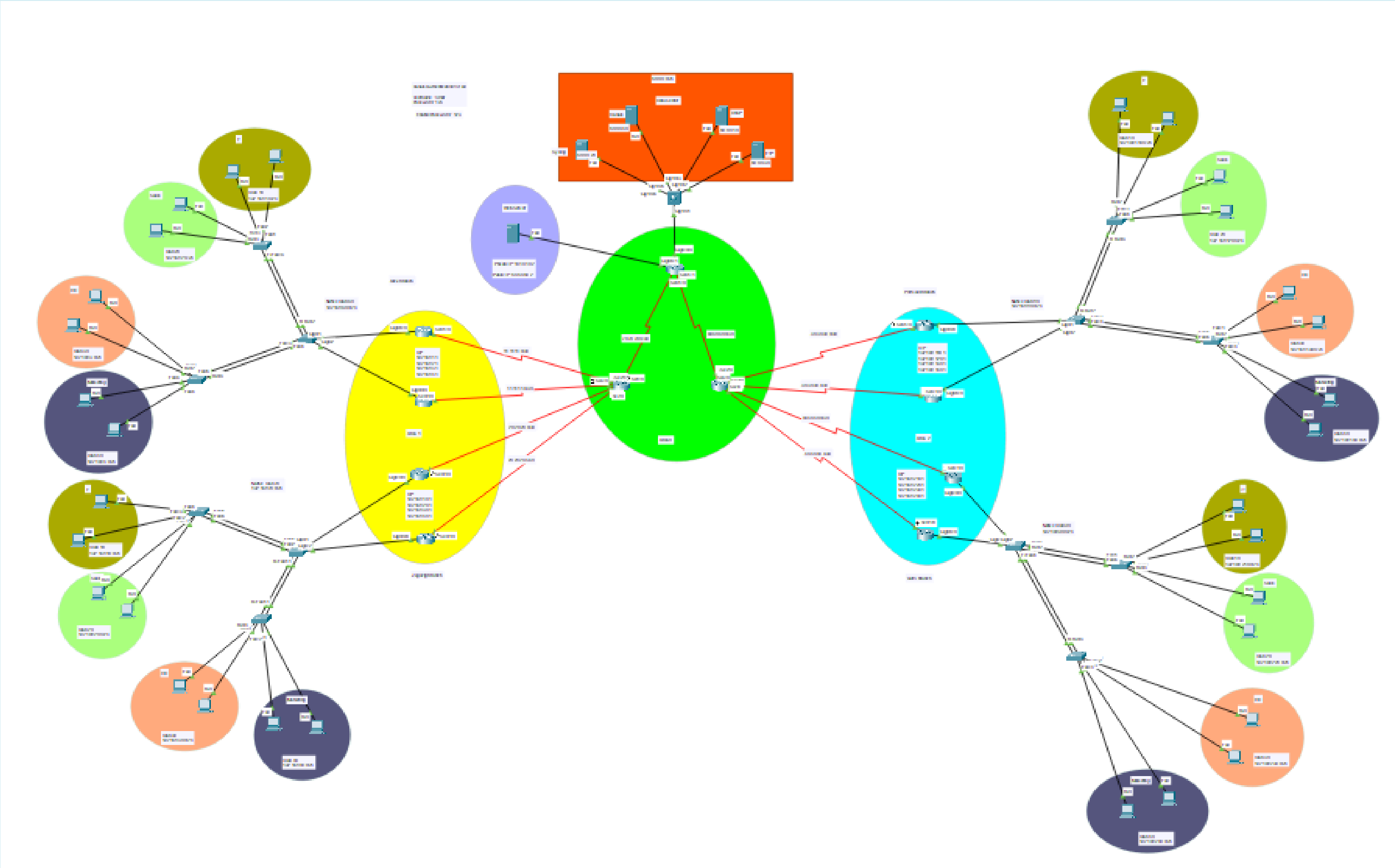
LINKING THE COMPANY'S BRANCHES TO EACH OTHER

PREPARED BY : OMAR'S TEAM

About the Project

Objective:
This project represents the comprehensive design, implementation, and documentation of a large-scale enterprise network infrastructure. The entire topology was built and simulated using Cisco Packet Tracer (v8.2.2), The design is architected to connect multiple branches and geographical areas (as depicted in the topology) while providing centralized services and secure, manageable access.
Core Technologies Implemented To achieve these objectives, a suite of industry-standard protocols and technologies was implemented, focusing on redundancy, security, and efficiency.

The Topology



Important Notes

This project has been fully implemented and tested. All devices have full end-to-end connectivity, and all services are operational as per the requirements.

- However, please be aware of the following behaviors observed within the Cisco Packet Tracer (v8.2.2) simulation environment:
- Initial Ping Timeouts: When testing connectivity for the first time (e.g., after a device boot or a link change), it is normal for the first one or two ping requests to Fail. This is expected behavior as protocols like ARP (Address Resolution Protocol) and STP (Spanning Tree Protocol) converge. All subsequent pings will be Successful.
- Configuration Integrity: All network devices (routers and switches) have been fully configured with all required features, including SSH for secure management. While all configurations are saved, the Packet Tracer application itself has occasionally shown instability. In some instances, the application may freeze or "hang," and upon restarting, it may fail to reload the previously saved configurations. This is a known artifact of the simulation tool and not a failure of the network design.

Scalable Routing (OSPF):

- Utilized OSPF (Open Shortest Path First) with a Multi-Area design. This ensures efficient routing, faster convergence, and a more scalable hierarchy by connecting all remote areas to the central Backbone (Area 0).

Network Segmentation (VLANs & ROAS):

- Implemented VLANs to logically segment the network into separate broadcast domains (e.g., by department). Router on a Stick (ROAS) was configured to enable and manage inter-VLAN communication efficiently.

High Availability & Redundancy:

- Ensured network resilience and service continuity through two key mechanisms:

HSRP (Hot Standby Router Protocol):

- Deployed to provide a redundant, virtual gateway for end devices, ensuring uninterrupted connectivity even if a primary router fails.

EtherChannel:

- Configured between critical switches to bundle multiple physical links into a single logical channel, providing increased bandwidth and link redundancy.

Centralized Services & Management:

DHCP Server:

- A stand-alone server was configured to dynamically assign IP addresses and other network configurations (Gateway, DNS) to end devices.

AAA Server:

- An Authentication, Authorization, and Accounting server was implemented for secure, centralized user authentication for network device management, with a local admin user as a backup.

FTP Server:

- Utilized to store periodic configuration backups from key network devices (routers and switches).

Security & Public Access:

Static NAT:

- Configured to publish an internal Web Server, making it accessible from the internet via a dedicated public IP address (50.50.50.2).

SSH (Secure Shell):

- Enabled on network devices to provide secure, encrypted remote access for administration, replacing the insecure Telnet protocol.

For AAA authentication

Username : omar
Password : 123

enable password : 123

Local authentication

Username : admin
Password : 123

Routing Protocol (OSPF)

OSPF Implementation

This network utilizes a hierarchical Multi-Area OSPF design to ensure scalable and efficient routing. The topology is built around a central Area 0 (Backbone Area), highlighted in green, which serves as the core of the network.

Area 1 (yellow) and Area 2 (cyan) are configured as standard non-backbone areas. As required by OSPF, both of these areas connect directly to the Area 0 backbone.
This design is crucial for scalability:

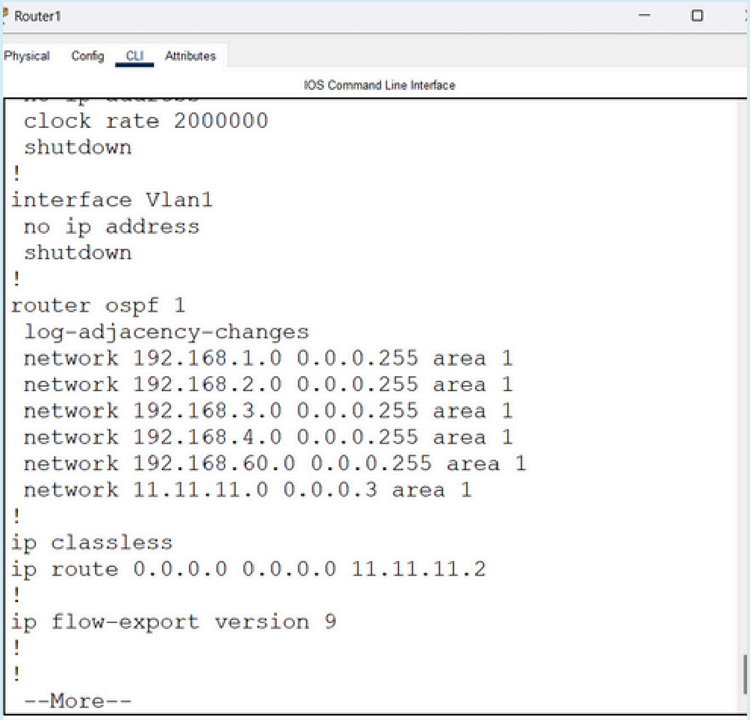
- Reduces Routing Table Size: Routers within Area 1 and Area 2 only need to maintain a full Link-State Database (LSDB) for their own area, receiving summarized routes from other areas.

- Faster Convergence: Any topology change within Area 1 is contained and does not force routers in Area 2 to recalculate their internal routes.

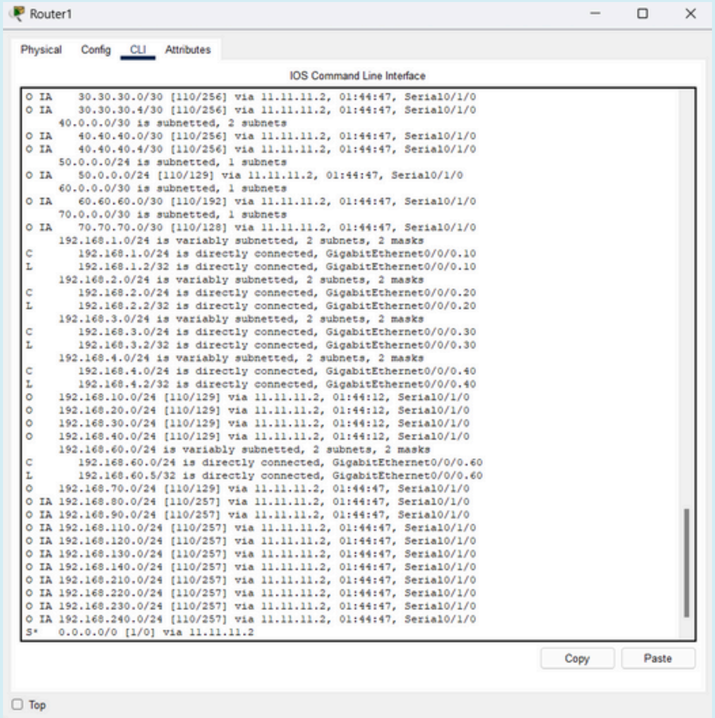
- Efficiency: Routers connecting the areas (Area Border Routers or ABRs) are responsible for filtering and summarizing routes, reducing CPU and memory overhead across the network.

There are the ospf configuration and the routing table from each router

Router 1 Area 1 Alex Branch

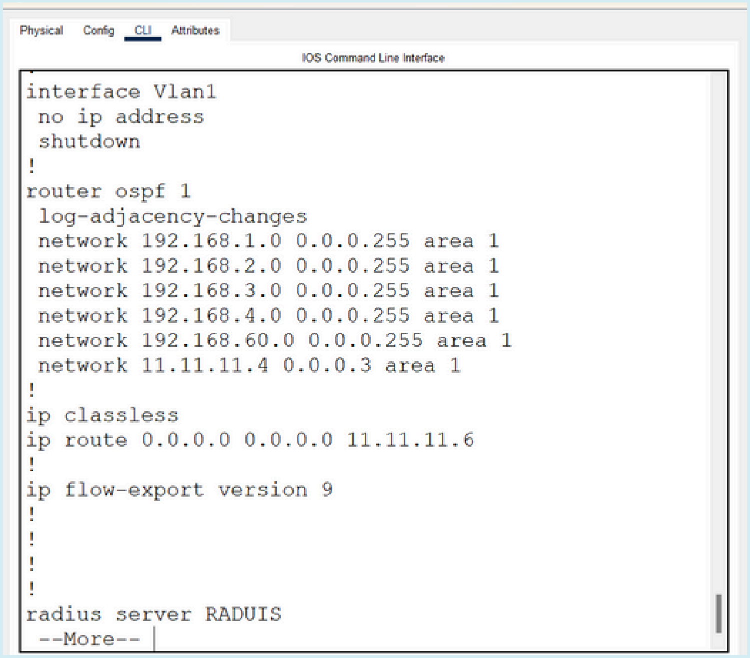


```
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 192.168.1.0 0.0.0.255 area 1
 network 192.168.2.0 0.0.0.255 area 1
 network 192.168.3.0 0.0.0.255 area 1
 network 192.168.4.0 0.0.0.255 area 1
 network 192.168.60.0 0.0.0.255 area 1
 network 11.11.11.0 0.0.0.3 area 1
!
ip classless
ip route 0.0.0.0 0.0.0.0 11.11.11.2
!
ip flow-export version 9
!
!
 --More--
```

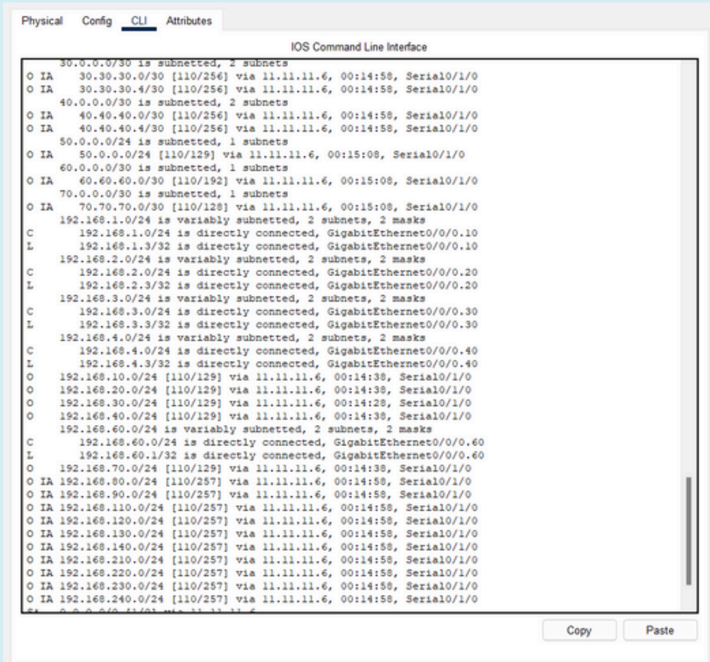


```
O IA    30.30.30.0/30 [110/256] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA    30.30.30.4/30 [110/256] via 11.11.11.2, 01:44:47, Serial0/1/0
     40.0.0.0/30 is subnetted, 2 subnets
O IA    40.40.40.0/30 [110/256] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA    40.40.40.4/30 [110/256] via 11.11.11.2, 01:44:47, Serial0/1/0
     50.0.0.0/24 is subnetted, 1 subnets
O IA    50.0.0.0/24 [110/129] via 11.11.11.2, 01:44:47, Serial0/1/0
     60.0.0.0/30 is subnetted, 1 subnets
O IA    60.60.60.0/30 [110/192] via 11.11.11.2, 01:44:47, Serial0/1/0
     70.0.0.0/30 is subnetted, 1 subnets
O IA    70.70.70.0/30 [110/128] via 11.11.11.2, 01:44:47, Serial0/1/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.1.2/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.2.2/32 is directly connected, GigabitEthernet0/0/0.20
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0/0.30
L       192.168.3.2/32 is directly connected, GigabitEthernet0/0/0.30
     192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.4.0/24 is directly connected, GigabitEthernet0/0/0.40
L       192.168.4.2/32 is directly connected, GigabitEthernet0/0/0.40
O    192.168.10.0/24 [110/129] via 11.11.11.2, 01:44:12, Serial0/1/0
O    192.168.20.0/24 [110/129] via 11.11.11.2, 01:44:12, Serial0/1/0
O    192.168.30.0/24 [110/129] via 11.11.11.2, 01:44:12, Serial0/1/0
O    192.168.40.0/24 [110/129] via 11.11.11.2, 01:44:12, Serial0/1/0
     192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.60.0/24 is directly connected, GigabitEthernet0/0/0.60
L       192.168.60.5/32 is directly connected, GigabitEthernet0/0/0.60
O    192.168.70.0/24 [110/129] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA 192.168.80.0/24 [110/257] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA 192.168.90.0/24 [110/257] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA 192.168.110.0/24 [110/257] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA 192.168.120.0/24 [110/257] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA 192.168.130.0/24 [110/257] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA 192.168.140.0/24 [110/257] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA 192.168.210.0/24 [110/257] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA 192.168.220.0/24 [110/257] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA 192.168.230.0/24 [110/257] via 11.11.11.2, 01:44:47, Serial0/1/0
O IA 192.168.240.0/24 [110/257] via 11.11.11.2, 01:44:47, Serial0/1/0
S*   0.0.0.0/0 [1/0] via 11.11.11.2
```

Router 2 Area 1 Alex Branch

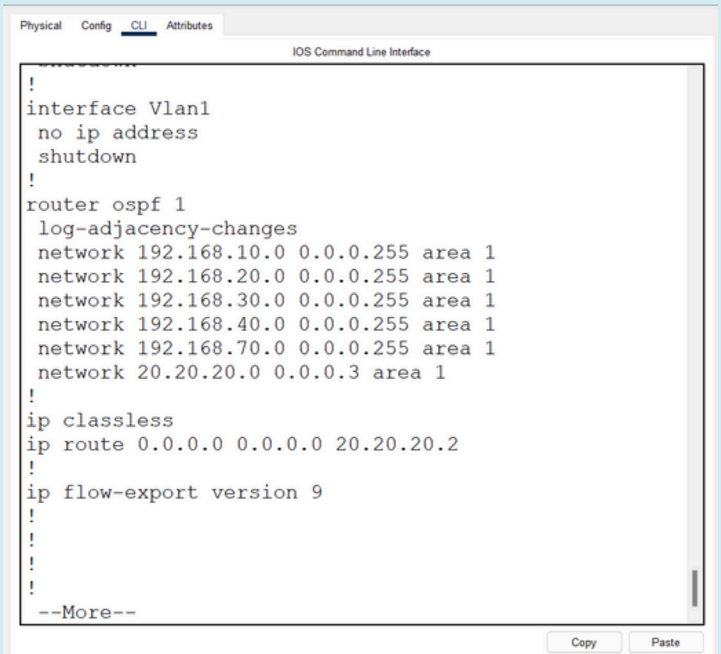


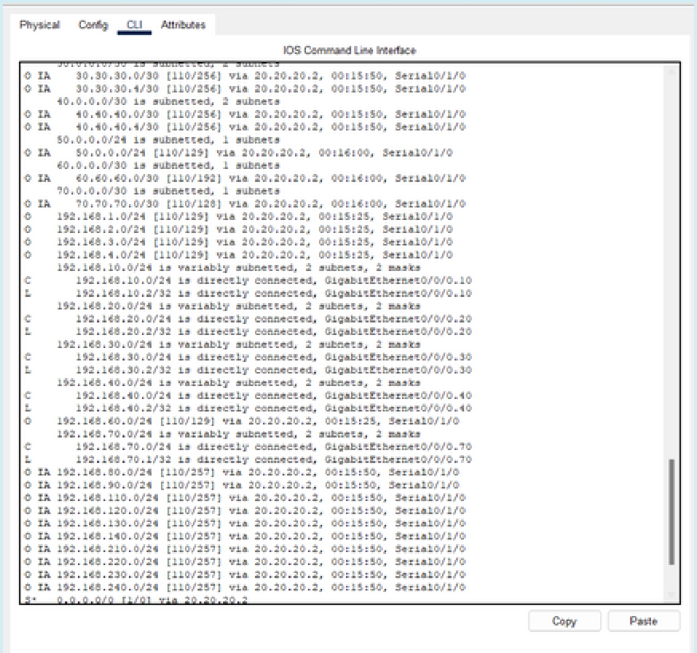
```
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 192.168.1.0 0.0.0.255 area 1
 network 192.168.2.0 0.0.0.255 area 1
 network 192.168.3.0 0.0.0.255 area 1
 network 192.168.4.0 0.0.0.255 area 1
 network 192.168.60.0 0.0.0.255 area 1
 network 11.11.11.4 0.0.0.3 area 1
!
ip classless
ip route 0.0.0.0 0.0.0.0 11.11.11.6
!
ip flow-export version 9
!
!
!
!
radius server RADUIS
 --More--
```



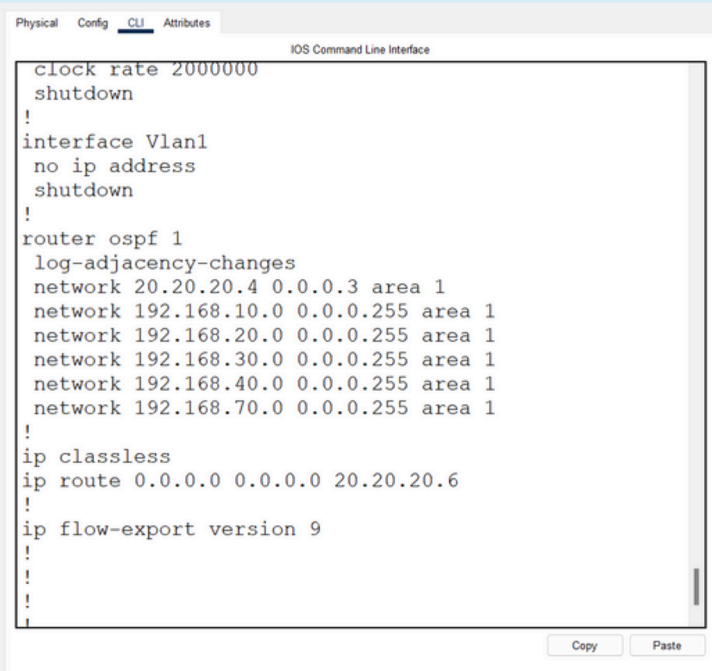
```
     30.0.0.0/30 is subnetted, 2 subnets
O IA    30.30.30.0/30 [110/256] via 11.11.11.6, 00:14:58, Serial0/1/0
O IA    30.30.30.4/30 [110/256] via 11.11.11.6, 00:14:58, Serial0/1/0
     40.0.0.0/30 is subnetted, 2 subnets
O IA    40.40.40.0/30 [110/256] via 11.11.11.6, 00:14:58, Serial0/1/0
O IA    40.40.40.4/30 [110/256] via 11.11.11.6, 00:14:58, Serial0/1/0
     50.0.0.0/24 is subnetted, 1 subnets
O IA    50.0.0.0/24 [110/129] via 11.11.11.6, 00:15:08, Serial0/1/0
     60.0.0.0/30 is subnetted, 1 subnets
O IA    60.60.60.0/30 [110/192] via 11.11.11.6, 00:15:08, Serial0/1/0
     70.0.0.0/30 is subnetted, 1 subnets
O IA    70.70.70.0/30 [110/128] via 11.11.11.6, 00:15:08, Serial0/1/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.1.3/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.2.3/32 is directly connected, GigabitEthernet0/0/0.20
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0/0.30
L       192.168.3.3/32 is directly connected, GigabitEthernet0/0/0.30
     192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.4.0/24 is directly connected, GigabitEthernet0/0/0.40
L       192.168.4.3/32 is directly connected, GigabitEthernet0/0/0.40
O    192.168.10.0/24 [110/129] via 11.11.11.6, 00:14:38, Serial0/1/0
O    192.168.20.0/24 [110/129] via 11.11.11.6, 00:14:38, Serial0/1/0
O    192.168.30.0/24 [110/129] via 11.11.11.6, 00:14:28, Serial0/1/0
O    192.168.40.0/24 [110/129] via 11.11.11.6, 00:14:38, Serial0/1/0
     192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.60.0/24 is directly connected, GigabitEthernet0/0/0.60
L       192.168.60.1/32 is directly connected, GigabitEthernet0/0/0.60
O    192.168.70.0/24 [110/129] via 11.11.11.6, 00:14:38, Serial0/1/0
O IA 192.168.80.0/24 [110/257] via 11.11.11.6, 00:14:58, Serial0/1/0
O IA 192.168.90.0/24 [110/257] via 11.11.11.6, 00:14:58, Serial0/1/0
O IA 192.168.110.0/24 [110/257] via 11.11.11.6, 00:14:58, Serial0/1/0
O IA 192.168.120.0/24 [110/257] via 11.11.11.6, 00:14:58, Serial0/1/0
O IA 192.168.130.0/24 [110/257] via 11.11.11.6, 00:14:58, Serial0/1/0
O IA 192.168.140.0/24 [110/257] via 11.11.11.6, 00:14:58, Serial0/1/0
O IA 192.168.210.0/24 [110/257] via 11.11.11.6, 00:14:58, Serial0/1/0
O IA 192.168.220.0/24 [110/257] via 11.11.11.6, 00:14:58, Serial0/1/0
O IA 192.168.230.0/24 [110/257] via 11.11.11.6, 00:14:58, Serial0/1/0
O IA 192.168.240.0/24 [110/257] via 11.11.11.6, 00:14:58, Serial0/1/0
```

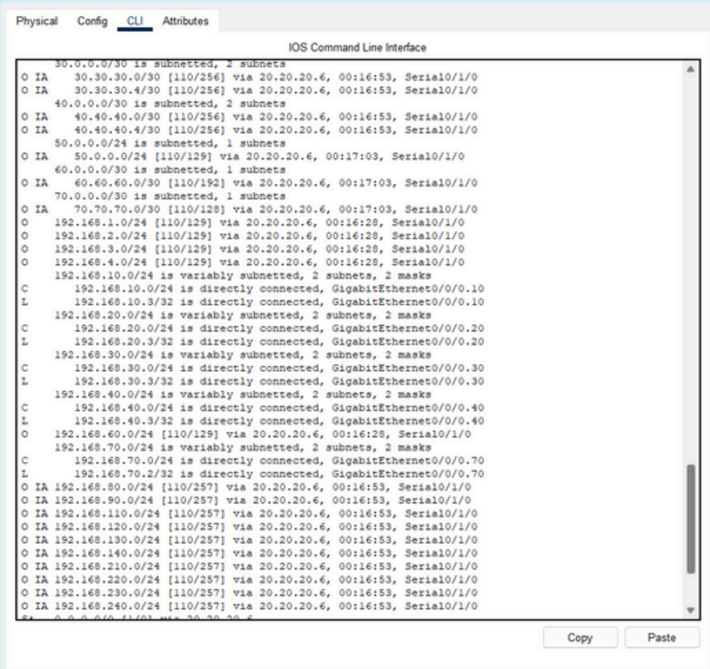
Router 1 Area 1 Zagazig Branch



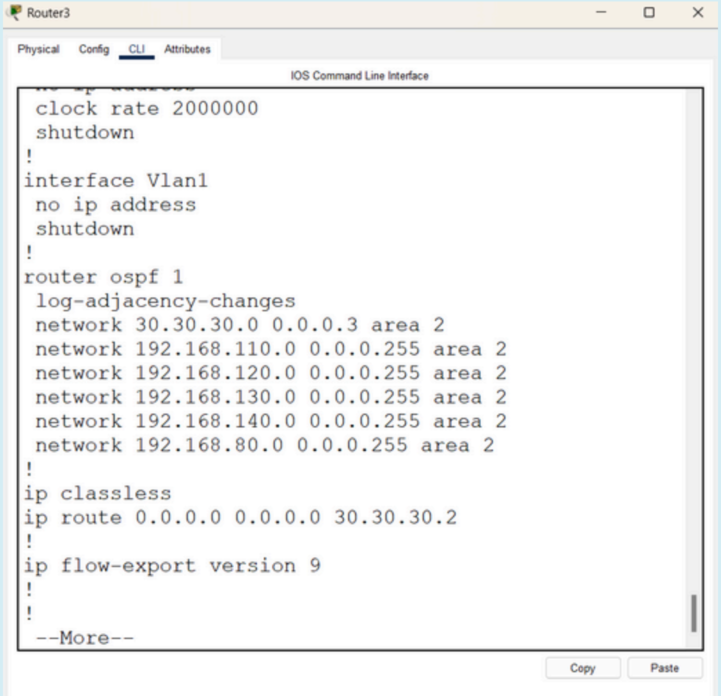


Router 2 Area 1 Zagazig Branch

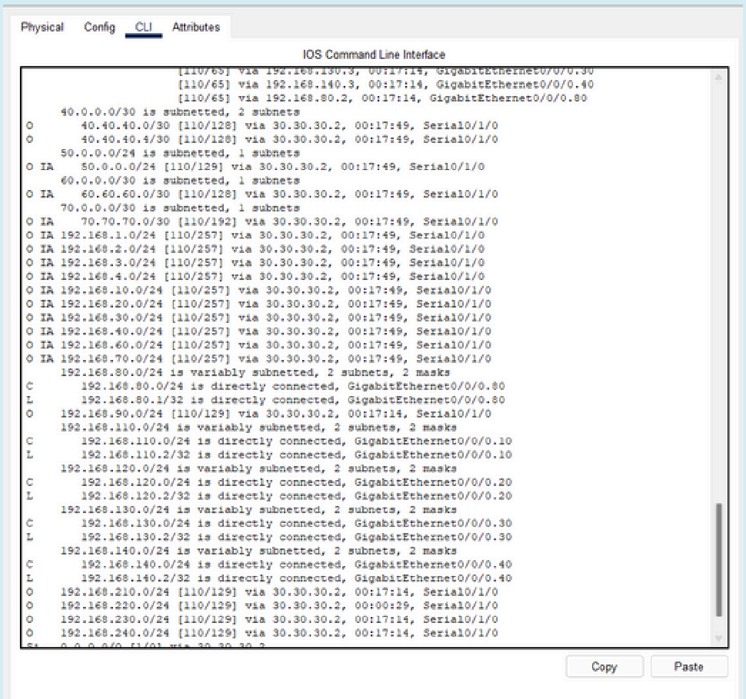




Router 1 Area 2 Port said Branch

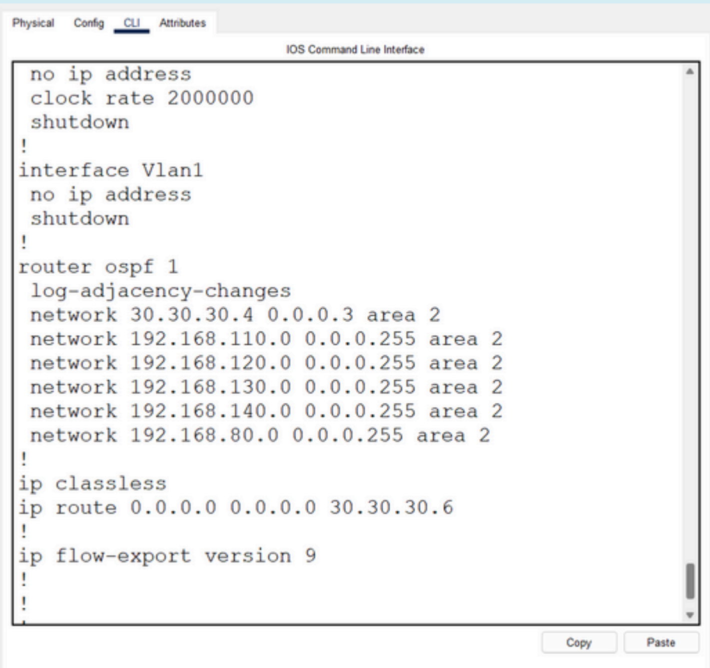


```
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 30.30.30.0 0.0.0.3 area 2
 network 192.168.110.0 0.0.0.255 area 2
 network 192.168.120.0 0.0.0.255 area 2
 network 192.168.130.0 0.0.0.255 area 2
 network 192.168.140.0 0.0.0.255 area 2
 network 192.168.80.0 0.0.0.255 area 2
!
ip classless
ip route 0.0.0.0 0.0.0.0 30.30.30.2
!
ip flow-export version 9
!
!
 --More--
```

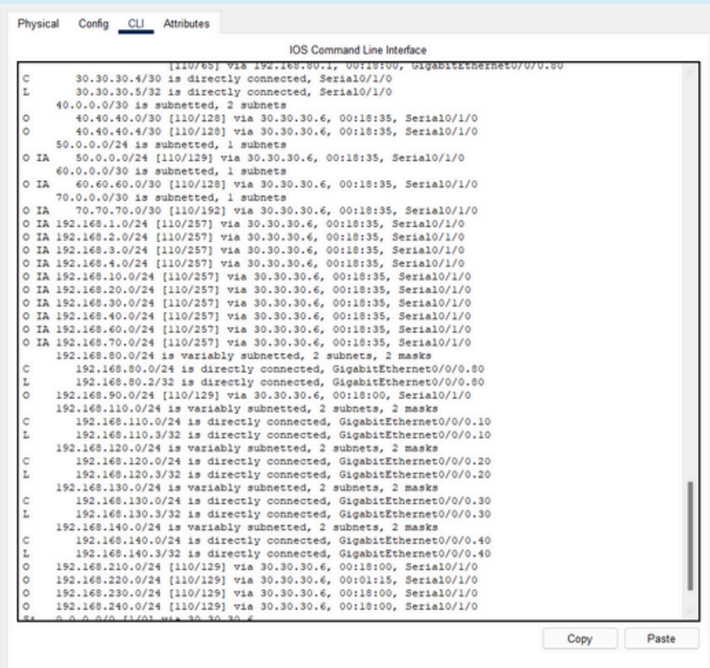


```
                    [110/65] via 192.168.140.3, 00:17:14, GigabitEthernet0/0/0.40
                    [110/65] via 192.168.80.2, 00:17:14, GigabitEthernet0/0/0.80
     40.0.0.0/30 is subnetted, 2 subnets
O       40.40.40.0/30 [110/128] via 30.30.30.2, 00:17:49, Serial0/1/0
O       40.40.40.4/30 [110/128] via 30.30.30.2, 00:17:49, Serial0/1/0
     50.0.0.0/24 is subnetted, 1 subnets
O IA    50.0.0.0/24 [110/129] via 30.30.30.2, 00:17:49, Serial0/1/0
     60.0.0.0/30 is subnetted, 1 subnets
O IA    60.60.60.0/30 [110/128] via 30.30.30.2, 00:17:49, Serial0/1/0
     70.0.0.0/30 is subnetted, 1 subnets
O IA    70.70.70.0/30 [110/192] via 30.30.30.2, 00:17:49, Serial0/1/0
O IA 192.168.1.0/24 [110/257] via 30.30.30.2, 00:17:49, Serial0/1/0
O IA 192.168.2.0/24 [110/257] via 30.30.30.2, 00:17:49, Serial0/1/0
O IA 192.168.3.0/24 [110/257] via 30.30.30.2, 00:17:49, Serial0/1/0
O IA 192.168.4.0/24 [110/257] via 30.30.30.2, 00:17:49, Serial0/1/0
O IA 192.168.10.0/24 [110/257] via 30.30.30.2, 00:17:49, Serial0/1/0
O IA 192.168.20.0/24 [110/257] via 30.30.30.2, 00:17:49, Serial0/1/0
O IA 192.168.30.0/24 [110/257] via 30.30.30.2, 00:17:49, Serial0/1/0
O IA 192.168.40.0/24 [110/257] via 30.30.30.2, 00:17:49, Serial0/1/0
O IA 192.168.60.0/24 [110/257] via 30.30.30.2, 00:17:49, Serial0/1/0
O IA 192.168.70.0/24 [110/257] via 30.30.30.2, 00:17:49, Serial0/1/0
     192.168.80.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.80.0/24 is directly connected, GigabitEthernet0/0/0.80
L       192.168.80.1/32 is directly connected, GigabitEthernet0/0/0.80
O    192.168.90.0/24 [110/129] via 30.30.30.2, 00:17:14, Serial0/1/0
     192.168.110.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.110.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.110.2/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.120.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.120.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.120.2/32 is directly connected, GigabitEthernet0/0/0.20
     192.168.130.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.130.0/24 is directly connected, GigabitEthernet0/0/0.30
L       192.168.130.2/32 is directly connected, GigabitEthernet0/0/0.30
     192.168.140.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.140.0/24 is directly connected, GigabitEthernet0/0/0.40
L       192.168.140.2/32 is directly connected, GigabitEthernet0/0/0.40
O    192.168.210.0/24 [110/129] via 30.30.30.2, 00:17:14, Serial0/1/0
O    192.168.220.0/24 [110/129] via 30.30.30.2, 00:00:29, Serial0/1/0
O    192.168.230.0/24 [110/129] via 30.30.30.2, 00:17:14, Serial0/1/0
O    192.168.240.0/24 [110/129] via 30.30.30.2, 00:17:14, Serial0/1/0
```

Router 2 Area 2 Port said Branch



```
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 30.30.30.4 0.0.0.3 area 2
 network 192.168.110.0 0.0.0.255 area 2
 network 192.168.120.0 0.0.0.255 area 2
 network 192.168.130.0 0.0.0.255 area 2
 network 192.168.140.0 0.0.0.255 area 2
 network 192.168.80.0 0.0.0.255 area 2
!
ip classless
ip route 0.0.0.0 0.0.0.0 30.30.30.6
!
ip flow-export version 9
!
!
```



```
C       30.30.30.4/30 is directly connected, Serial0/1/0
L       30.30.30.5/32 is directly connected, Serial0/1/0
     40.0.0.0/30 is subnetted, 2 subnets
O       40.40.40.0/30 [110/128] via 30.30.30.6, 00:18:35, Serial0/1/0
O       40.40.40.4/30 [110/128] via 30.30.30.6, 00:18:35, Serial0/1/0
     50.0.0.0/24 is subnetted, 1 subnets
O IA    50.0.0.0/24 [110/129] via 30.30.30.6, 00:18:35, Serial0/1/0
     60.0.0.0/30 is subnetted, 1 subnets
O IA    60.60.60.0/30 [110/128] via 30.30.30.6, 00:18:35, Serial0/1/0
     70.0.0.0/30 is subnetted, 1 subnets
O IA    70.70.70.0/30 [110/192] via 30.30.30.6, 00:18:35, Serial0/1/0
O IA 192.168.1.0/24 [110/257] via 30.30.30.6, 00:18:35, Serial0/1/0
O IA 192.168.2.0/24 [110/257] via 30.30.30.6, 00:18:35, Serial0/1/0
O IA 192.168.3.0/24 [110/257] via 30.30.30.6, 00:18:35, Serial0/1/0
O IA 192.168.4.0/24 [110/257] via 30.30.30.6, 00:18:35, Serial0/1/0
O IA 192.168.10.0/24 [110/257] via 30.30.30.6, 00:18:35, Serial0/1/0
O IA 192.168.20.0/24 [110/257] via 30.30.30.6, 00:18:35, Serial0/1/0
O IA 192.168.30.0/24 [110/257] via 30.30.30.6, 00:18:35, Serial0/1/0
O IA 192.168.40.0/24 [110/257] via 30.30.30.6, 00:18:35, Serial0/1/0
O IA 192.168.60.0/24 [110/257] via 30.30.30.6, 00:18:35, Serial0/1/0
O IA 192.168.70.0/24 [110/257] via 30.30.30.6, 00:18:35, Serial0/1/0
     192.168.80.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.80.0/24 is directly connected, GigabitEthernet0/0/0.80
L       192.168.80.2/32 is directly connected, GigabitEthernet0/0/0.80
O    192.168.90.0/24 [110/129] via 30.30.30.6, 00:18:00, Serial0/1/0
     192.168.110.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.110.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.110.3/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.120.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.120.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.120.3/32 is directly connected, GigabitEthernet0/0/0.20
     192.168.130.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.130.0/24 is directly connected, GigabitEthernet0/0/0.30
L       192.168.130.3/32 is directly connected, GigabitEthernet0/0/0.30
     192.168.140.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.140.0/24 is directly connected, GigabitEthernet0/0/0.40
L       192.168.140.3/32 is directly connected, GigabitEthernet0/0/0.40
O    192.168.210.0/24 [110/129] via 30.30.30.6, 00:18:00, Serial0/1/0
O    192.168.220.0/24 [110/129] via 30.30.30.6, 00:01:15, Serial0/1/0
O    192.168.230.0/24 [110/129] via 30.30.30.6, 00:18:00, Serial0/1/0
O    192.168.240.0/24 [110/129] via 30.30.30.6, 00:18:00, Serial0/1/0
```

Router 1 Area 2 Cairo Branch

Physical Config CLI Attributes

IOS Command Line Interface

```
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 40.40.40.0 0.0.0.3 area 2
 network 192.168.210.0 0.0.0.255 area 2
 network 192.168.220.0 0.0.0.255 area 2
 network 192.168.230.0 0.0.0.255 area 2
 network 192.168.240.0 0.0.0.255 area 2
 network 192.168.90.0 0.0.0.255 area 2
!
ip classless
ip route 0.0.0.0 0.0.0.0 40.40.40.2
!
ip flow-export version 9
!
!
 --More--
```

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Physical Config CLI Attributes

IOS Command Line Interface

```
O       40.40.40.4/30 [110/65] via 192.168.210.3, 00:00:57, GigabitEthernet0/0/0.10
                      [110/65] via 192.168.220.3, 00:00:57, GigabitEthernet0/0/0.20
                      [110/65] via 192.168.230.3, 00:00:57, GigabitEthernet0/0/0.30
                      [110/65] via 192.168.240.3, 00:00:57, GigabitEthernet0/0/0.40
                      [110/65] via 192.168.90.2, 00:00:57, GigabitEthernet0/0/0.90
     50.0.0.0/24 is subnetted, 1 subnets
O IA    50.0.0.0/24 [110/129] via 40.40.40.2, 00:08:16, Serial0/1/0
     60.0.0.0/30 is subnetted, 1 subnets
O IA    60.60.60.0/30 [110/128] via 40.40.40.2, 00:08:16, Serial0/1/0
     70.0.0.0/30 is subnetted, 1 subnets
O IA    70.70.70.0/30 [110/192] via 40.40.40.2, 00:08:16, Serial0/1/0
O IA 192.168.1.0/24 [110/257] via 40.40.40.2, 00:08:16, Serial0/1/0
O IA 192.168.2.0/24 [110/257] via 40.40.40.2, 00:08:16, Serial0/1/0
O IA 192.168.3.0/24 [110/257] via 40.40.40.2, 00:08:16, Serial0/1/0
O IA 192.168.4.0/24 [110/257] via 40.40.40.2, 00:08:16, Serial0/1/0
O IA 192.168.10.0/24 [110/257] via 40.40.40.2, 00:08:16, Serial0/1/0
O IA 192.168.20.0/24 [110/257] via 40.40.40.2, 00:08:16, Serial0/1/0
O IA 192.168.30.0/24 [110/257] via 40.40.40.2, 00:08:16, Serial0/1/0
O IA 192.168.40.0/24 [110/257] via 40.40.40.2, 00:08:16, Serial0/1/0
O IA 192.168.60.0/24 [110/257] via 40.40.40.2, 00:08:16, Serial0/1/0
O IA 192.168.70.0/24 [110/257] via 40.40.40.2, 00:08:16, Serial0/1/0
O    192.168.80.0/24 [110/129] via 40.40.40.2, 00:08:16, Serial0/1/0
     192.168.90.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.90.0/24 is directly connected, GigabitEthernet0/0/0.90
L       192.168.90.1/32 is directly connected, GigabitEthernet0/0/0.90
O    192.168.110.0/24 [110/129] via 40.40.40.2, 00:08:16, Serial0/1/0
O    192.168.120.0/24 [110/129] via 40.40.40.2, 00:08:16, Serial0/1/0
O    192.168.130.0/24 [110/129] via 40.40.40.2, 00:08:16, Serial0/1/0
O    192.168.140.0/24 [110/129] via 40.40.40.2, 00:08:16, Serial0/1/0
     192.168.210.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.210.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.210.2/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.220.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.220.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.220.2/32 is directly connected, GigabitEthernet0/0/0.20
     192.168.230.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.230.0/24 is directly connected, GigabitEthernet0/0/0.30
L       192.168.230.2/32 is directly connected, GigabitEthernet0/0/0.30
     192.168.240.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.240.0/24 is directly connected, GigabitEthernet0/0/0.40
L       192.168.240.2/32 is directly connected, GigabitEthernet0/0/0.40
```

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Router 2 Area 2 Cairo Branch

Physical Config CLI Attributes

IOS Command Line Interface

```
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 40.40.40.4 0.0.0.3 area 2
 network 192.168.210.0 0.0.0.255 area 2
 network 192.168.220.0 0.0.0.255 area 2
 network 192.168.230.0 0.0.0.255 area 2
 network 192.168.240.0 0.0.0.255 area 2
 network 192.168.90.0 0.0.0.255 area 2
!
ip classless
ip route 0.0.0.0 0.0.0.0 40.40.40.6
!
ip flow-export version 9
!
!
!
```

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Physical Config CLI Attributes

IOS Command Line Interface

```
O       40.40.40.0/30 [110/65] via 192.168.210.2, 00:00:05, GigabitEthernet0/0/0.10
                      [110/65] via 192.168.230.2, 00:00:05, GigabitEthernet0/0/0.30
                      [110/65] via 192.168.240.2, 00:00:05, GigabitEthernet0/0/0.40
                      [110/65] via 192.168.90.1, 00:00:05, GigabitEthernet0/0/0.90
C       40.40.40.4/30 is directly connected, Serial0/1/0
L       40.40.40.5/32 is directly connected, Serial0/1/0
     50.0.0.0/24 is subnetted, 1 subnets
O IA    50.0.0.0/24 [110/129] via 40.40.40.6, 01:25:43, Serial0/1/0
     60.0.0.0/30 is subnetted, 1 subnets
O IA    60.60.60.0/30 [110/128] via 40.40.40.6, 01:25:43, Serial0/1/0
     70.0.0.0/30 is subnetted, 1 subnets
O IA    70.70.70.0/30 [110/192] via 40.40.40.6, 01:25:43, Serial0/1/0
O IA 192.168.1.0/24 [110/257] via 40.40.40.6, 01:25:43, Serial0/1/0
O IA 192.168.2.0/24 [110/257] via 40.40.40.6, 01:25:43, Serial0/1/0
O IA 192.168.3.0/24 [110/257] via 40.40.40.6, 01:25:43, Serial0/1/0
O IA 192.168.4.0/24 [110/257] via 40.40.40.6, 01:25:43, Serial0/1/0
O IA 192.168.10.0/24 [110/257] via 40.40.40.6, 01:25:43, Serial0/1/0
O IA 192.168.20.0/24 [110/257] via 40.40.40.6, 01:25:43, Serial0/1/0
O IA 192.168.30.0/24 [110/257] via 40.40.40.6, 01:25:43, Serial0/1/0
O IA 192.168.40.0/24 [110/257] via 40.40.40.6, 01:25:43, Serial0/1/0
O IA 192.168.60.0/24 [110/257] via 40.40.40.6, 01:25:43, Serial0/1/0
O IA 192.168.70.0/24 [110/257] via 40.40.40.6, 01:25:43, Serial0/1/0
O    192.168.80.0/24 [110/129] via 40.40.40.6, 00:13:45, Serial0/1/0
     192.168.90.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.90.0/24 is directly connected, GigabitEthernet0/0/0.90
L       192.168.90.2/32 is directly connected, GigabitEthernet0/0/0.90
O    192.168.110.0/24 [110/129] via 40.40.40.6, 00:13:45, Serial0/1/0
O    192.168.120.0/24 [110/129] via 40.40.40.6, 00:13:45, Serial0/1/0
O    192.168.130.0/24 [110/129] via 40.40.40.6, 00:13:45, Serial0/1/0
O    192.168.140.0/24 [110/129] via 40.40.40.6, 00:13:45, Serial0/1/0
     192.168.210.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.210.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.210.3/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.220.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.220.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.220.3/32 is directly connected, GigabitEthernet0/0/0.20
     192.168.230.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.230.0/24 is directly connected, GigabitEthernet0/0/0.30
L       192.168.230.3/32 is directly connected, GigabitEthernet0/0/0.30
     192.168.240.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.240.0/24 is directly connected, GigabitEthernet0/0/0.40
L       192.168.240.3/32 is directly connected, GigabitEthernet0/0/0.40
```

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Network Segmentation (VLANs & ROAS)

- This infrastructure is designed to support four main corporate branches. To enhance security, manage broadcast traffic, and logically organize users, each branch is segmented using VLANs to create separate networks for its four primary departments:

- IT
- Sales
- HR
- Marketing

Inter-VLAN Routing (ROAS) :

- Communication between these different departmental VLANs is enabled using the 'Router on a Stick' (ROAS) model. This is achieved by configuring 802.1q sub-interfaces on the main branch router, one for each departmental VLAN.

Native VLAN Configuration :

- A dedicated Native VLAN is also configured for each branch's trunk links This VLAN is assigned its own unique subnet and, like the other departments, is configured as a routed sub-interface on the router.

Add Routing Integration (OSPF) :

- To ensure full, enterprise-wide connectivity, all these routed networks—including all departmental VLAN sub-interfaces and the Native VLAN sub-interface—are advertised into the OSPF routing protocol. This allows all branches to dynamically learn and maintain routes to every specific subnet across the entire organization.

Each router connected to switches has this configuration based on the IPs of gateways of its VLAN

IOS Command Line Interface

```
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.1.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.20
 encapsulation dot1Q 20
 ip address 192.168.2.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.2.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.30
 encapsulation dot1Q 30
 ip address 192.168.3.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.3.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.40
 encapsulation dot1Q 40
 ip address 192.168.4.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.4.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.60
 encapsulation dot1Q 60 native
 ip address 192.168.60.1 255.255.255.0
!
interface GigabitEthernet0/0/1
 --More--
```

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High Availability and Redundancy

HSRP (Hot Standby Router Protocol):

- Default gateway redundancy is achieved by deploying HSRP on the core routers. These routers are configured as an HSRP group, presenting a single virtual IP address to the network, which is used as the default gateway by the branch routers (ABRs).

EtherChannel (Link Aggregation):

- To provide link redundancy and increased bandwidth, EtherChannel is configured between the switches.
- This implementation bundles multiple physical links into a single logical connection. This design provides two major benefits:

- **Fault Tolerance:** If one physical link within the bundle fails, traffic is automatically re-distributed to the remaining links with no service interruption.
- **Increased Bandwidth:** The bandwidth of all active links is aggregated, creating a high-capacity trunk that can handle the significant traffic loads between the core devices.

HSRP configuration

IOS Command Line Interface

```
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.1.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.20
 encapsulation dot1Q 20
 ip address 192.168.2.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.2.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.30
 encapsulation dot1Q 30
 ip address 192.168.3.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.3.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.40
 encapsulation dot1Q 40
 ip address 192.168.4.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.4.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.60
 encapsulation dot1Q 60 native
 ip address 192.168.60.1 255.255.255.0
!
interface GigabitEthernet0/0/1
 --More--
```

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EtherChannel configuration

Physical Config CLI Attributes

IOS Command Line Interface

```
!
!
!
!
ip domain-name iti.com
!
username admin secret 5 $1$mERr$3HhIgMGBA/9qNmgzccuxv0
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface Port-channel1
 switchport trunk native vlan 60
 switchport mode trunk
!
interface Port-channel2
 switchport trunk native vlan 60
 switchport mode trunk
!
interface FastEthernet0/1
 switchport trunk native vlan 60
 switchport mode trunk
 channel-group 1 mode active
!
interface FastEthernet0/2
 switchport trunk native vlan 60
 switchport mode trunk
 channel-group 1 mode active
!
interface FastEthernet0/3
 switchport trunk native vlan 60
 switchport mode trunk
 channel-group 2 mode active
!
interface FastEthernet0/4
 switchport trunk native vlan 60
 switchport mode trunk
 channel-group 2 mode active
!
```

DHCP

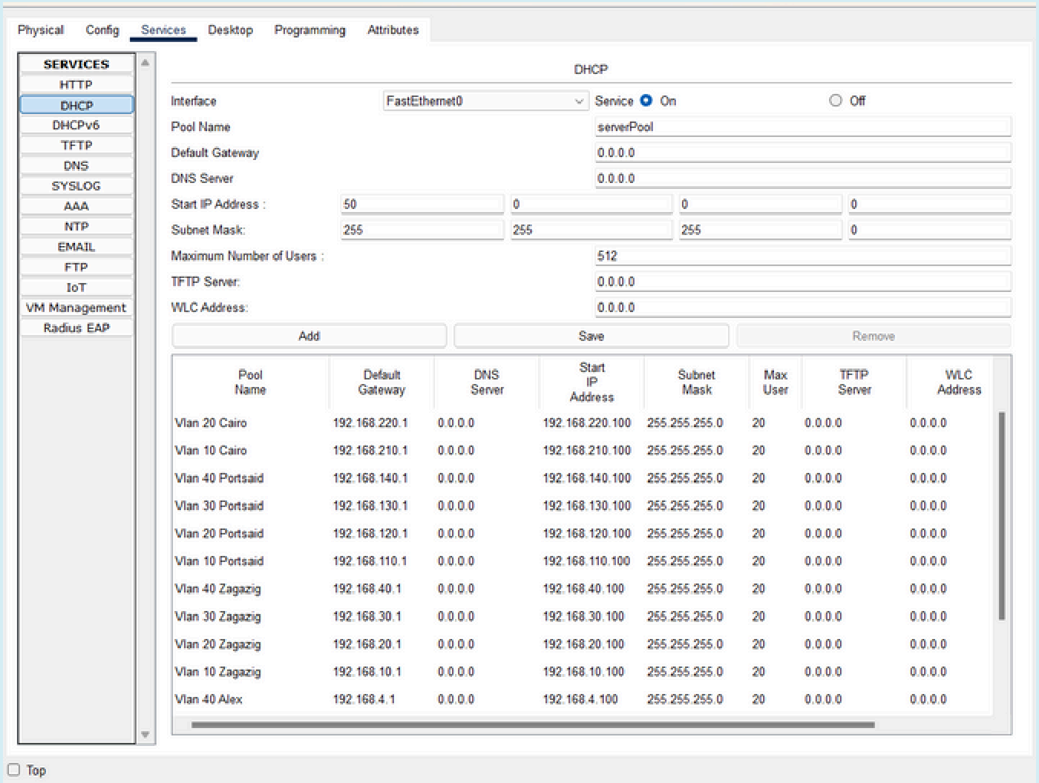
Centralized DHCP Services

- To streamline IP address management and simplify network administration, a centralized, stand-alone DHCP server is deployed.
- This server is strategically placed within the Backbone (Area 0), allowing it to provide DHCP services to all VLANs and subnets across the entire enterprise, including those in Area 1 and Area 2.

Implementation Details:

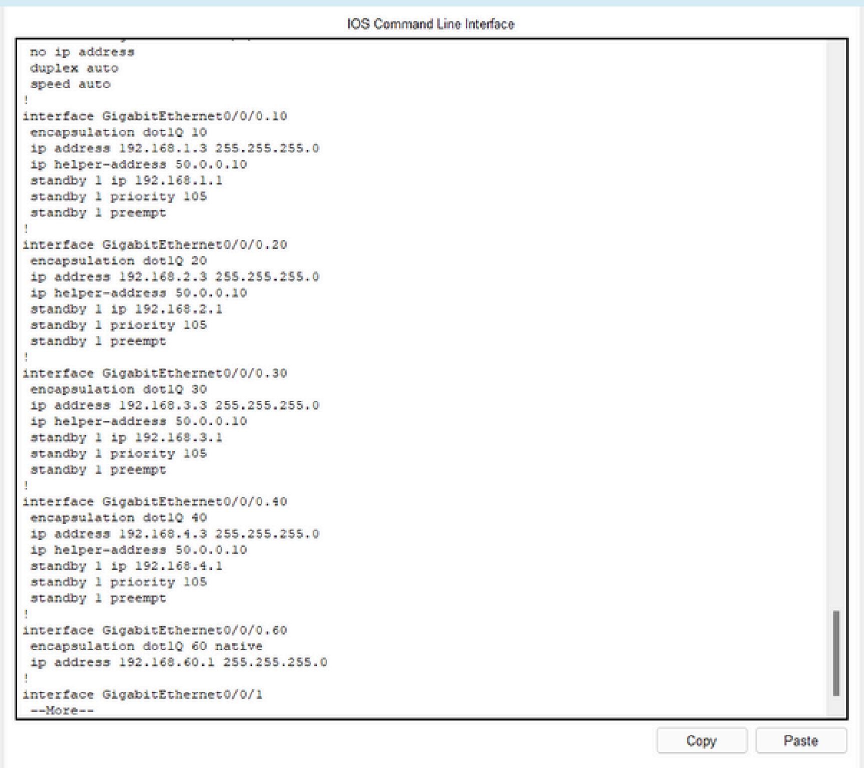
- Centralized Pools:
  The DHCP server is configured with a separate IP address pool for every single VLAN (e.g., IT, Sales, HR, Marketing) in all four branches.
- DHCP Relay:
  To allow broadcast-based DHCP requests from the branches to reach the centralized server (which is on a different subnet), the ip helper-address command is configured on each router's sub-interface (the default gateway for each VLAN).
- Function:
  When a client in any VLAN boots up, its DHCP request is forwarded by its local router (acting as a DHCP Relay Agent) directly to the DHCP server in Area 0. The server then assigns an appropriate IP address from the correct pool based on the request's source.

Pools on DHCP server



Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Vlan 20 Cairo	192.168.220.1	0.0.0.0	192.168.220.100	255.255.255.0	20	0.0.0.0	0.0.0.0
Vlan 10 Cairo	192.168.210.1	0.0.0.0	192.168.210.100	255.255.255.0	20	0.0.0.0	0.0.0.0
Vlan 40 Portsaid	192.168.140.1	0.0.0.0	192.168.140.100	255.255.255.0	20	0.0.0.0	0.0.0.0
Vlan 30 Portsaid	192.168.130.1	0.0.0.0	192.168.130.100	255.255.255.0	20	0.0.0.0	0.0.0.0
Vlan 20 Portsaid	192.168.120.1	0.0.0.0	192.168.120.100	255.255.255.0	20	0.0.0.0	0.0.0.0
Vlan 10 Portsaid	192.168.110.1	0.0.0.0	192.168.110.100	255.255.255.0	20	0.0.0.0	0.0.0.0
Vlan 40 Zagazig	192.168.40.1	0.0.0.0	192.168.40.100	255.255.255.0	20	0.0.0.0	0.0.0.0
Vlan 30 Zagazig	192.168.30.1	0.0.0.0	192.168.30.100	255.255.255.0	20	0.0.0.0	0.0.0.0
Vlan 20 Zagazig	192.168.20.1	0.0.0.0	192.168.20.100	255.255.255.0	20	0.0.0.0	0.0.0.0
Vlan 10 Zagazig	192.168.10.1	0.0.0.0	192.168.10.100	255.255.255.0	20	0.0.0.0	0.0.0.0
Vlan 40 Alex	192.168.4.1	0.0.0.0	192.168.4.100	255.255.255.0	20	0.0.0.0	0.0.0.0

DHCP Relay configuration in sub-interfaces




```
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.1.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.20
 encapsulation dot1Q 20
 ip address 192.168.2.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.2.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.30
 encapsulation dot1Q 30
 ip address 192.168.3.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.3.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.40
 encapsulation dot1Q 40
 ip address 192.168.4.3 255.255.255.0
 ip helper-address 50.0.0.10
 standby 1 ip 192.168.4.1
 standby 1 priority 105
 standby 1 preempt
!
interface GigabitEthernet0/0/0.60
 encapsulation dot1Q 60 native
 ip address 192.168.60.1 255.255.255.0
!
interface GigabitEthernet0/0/1
 --More--
```

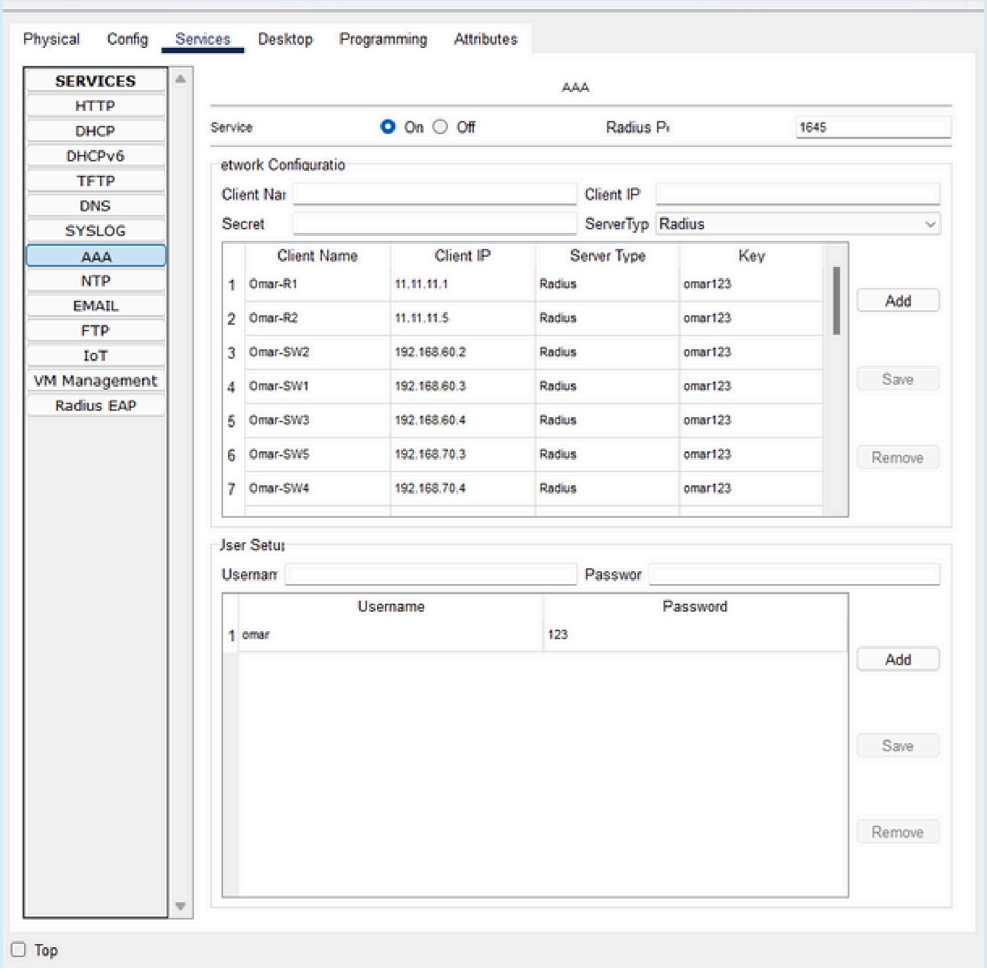
Security & Device Access (AAA with RADIUS)

- To enforce a robust and centralized security policy for device management, a central RADIUS server is used to manage administrator authentication.
- This approach requires all administrators to authenticate with a unique username and password validated by the RADIUS server, rather than relying on shared, device-specific passwords.

Implementation Scope:

- Routers (Full Enforcement): All routers are configured to use RADIUS authentication for both access methods:
- Console Access (line con 0): Secures direct, physical access.
- Remote Access (line vty): Secures all remote management sessions (e.g., SSH).
- Switches (Remote Enforcement): All switches are configured to use RADIUS authentication for Remote Access (line vty), ensuring any administrator attempting to manage the switch must first be validated by the RADIUS server.
- A local admin user account is also configured on each device. This local database serves as a critical fallback authentication method, ensuring that management access is still possible if the central RADIUS server becomes unreachable.

Configuration from radius server



Configuration from routers and switches

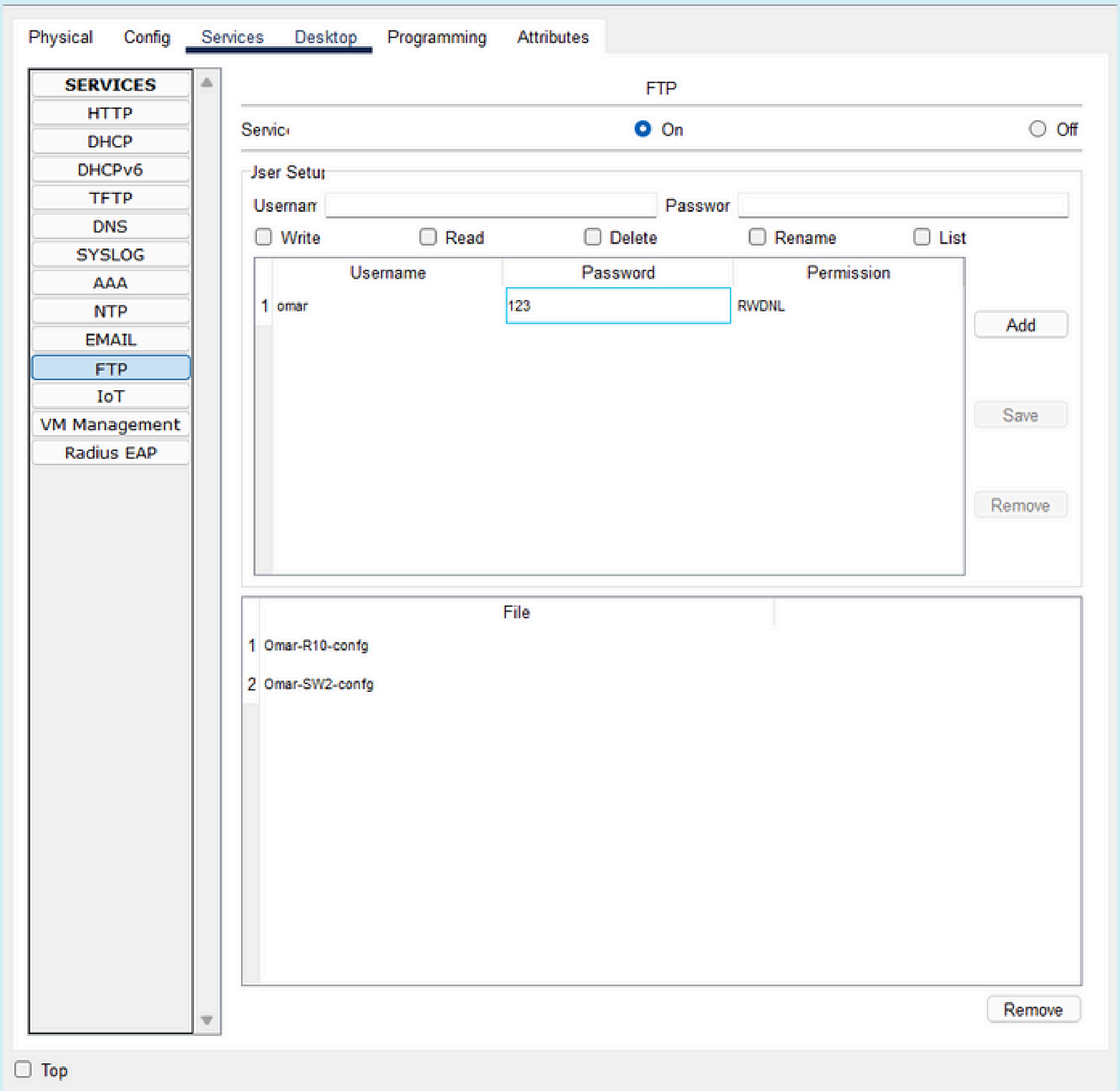
```
!
aaa new-model
!
aaa authentication login MY-LIST group radius local
!
```

```
radius server RADUIS
 address ipv4 50.0.0.40 auth-port 1645
 key omar123
radius server 50.0.0.40
 address ipv4 50.0.0.40 auth-port 1645
 key omar123
!
```

```
Omar-SW5(config)#aaa new-model
Omar-SW5(config)#aaa authentication login MY-LIST group radius local
Omar-SW5(config)#radius server RADIUS
Omar-SW5(config-radius-server)#address ipv4 50.0.0.40
Omar-SW5(config-radius-server)#key omar123
```

FTP Server

- Centralized File Services (FTP Server)
- A central FTP server is deployed within the network backbone (Area 0).
- This strategic placement ensures the server is reachable from all network devices across every branch and area. Its primary function is to serve as a central repository for storing configuration backups from all routers and switches, simplifying disaster recovery and device replacement.



Configuration from Router 10

```
ip ftp username omar
ip ftp password 123
```

```
Omar-R10#copy startup-config ftp
Omar-R10#copy startup-config ftp:
Address or name of remote host []? 50.0.0.30
Destination filename [Omar-R10-confg]?

Writing startup-config...
[OK - 1598 bytes]

1598 bytes copied in 0.019 secs (84000 bytes/sec)
Omar-R10#
```

Copy Paste

Configuration from Switch 2

```
hostname Omar-SW2
!
ip ftp username omar
ip ftp password 123
enable secret 5 $1$mERr$3HhIgMGBA/9qNmgzccuxv0
!
```

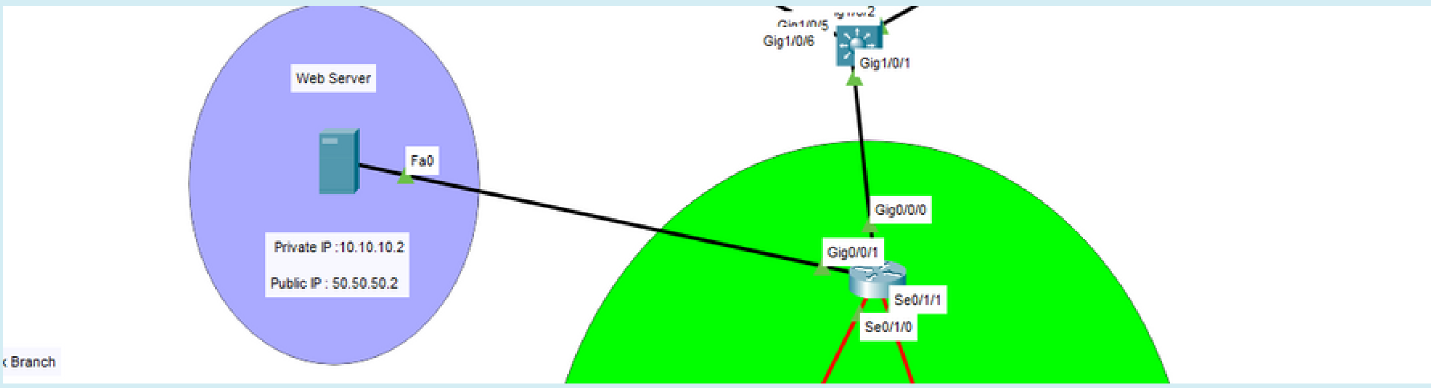
```
Omar-SW2#show flash
Directory of flash:/

    1  -rw-     4670455          <no date>  2960-lanbasek9-mz.150-2.SE4.bin
   13  -rw-        2075          <no date>  config.text
    2  -rw-        1096          <no date>  vlan.dat

64016384 bytes total (59342758 bytes free)
Omar-SW2#copy flash: ftp:
Source filename []? 2960-lanbasek9-mz.150-2.SE4.bin
Address or name of remote host []? 50.0.0.30
Destination filename [2960-lanbasek9-mz.150-2.SE4.bin]? IOS-SW2-Backup

Writing 2960-lanbasek9-mz.150-2.SE4.bin...
[OK - 4670455 bytes]

4670455 bytes copied in 9.132 secs (511000 bytes/sec)
Omar-SW2#
```

Copy Paste

Static NAT

Public Service Access (Static NAT) :

- To make internal resources available to the public internet, a Web Server is hosted within the network.
- This server resides on a secure, private network segment with the internal IP address 10.10.10.2. To allow external users to access this server, a Static NAT (Network Address Translation) mapping has been configured on the edge router.
- This configuration creates a permanent, one-to-one translation, mapping the public IP address 50.50.50.2 to the server's private IP address. As a result, any external traffic destined for 50.50.50.2 is automatically translated and forwarded to the internal Web Server at 10.10.10.2, effectively publishing the service to the outside world while protecting the server's true location on the private network

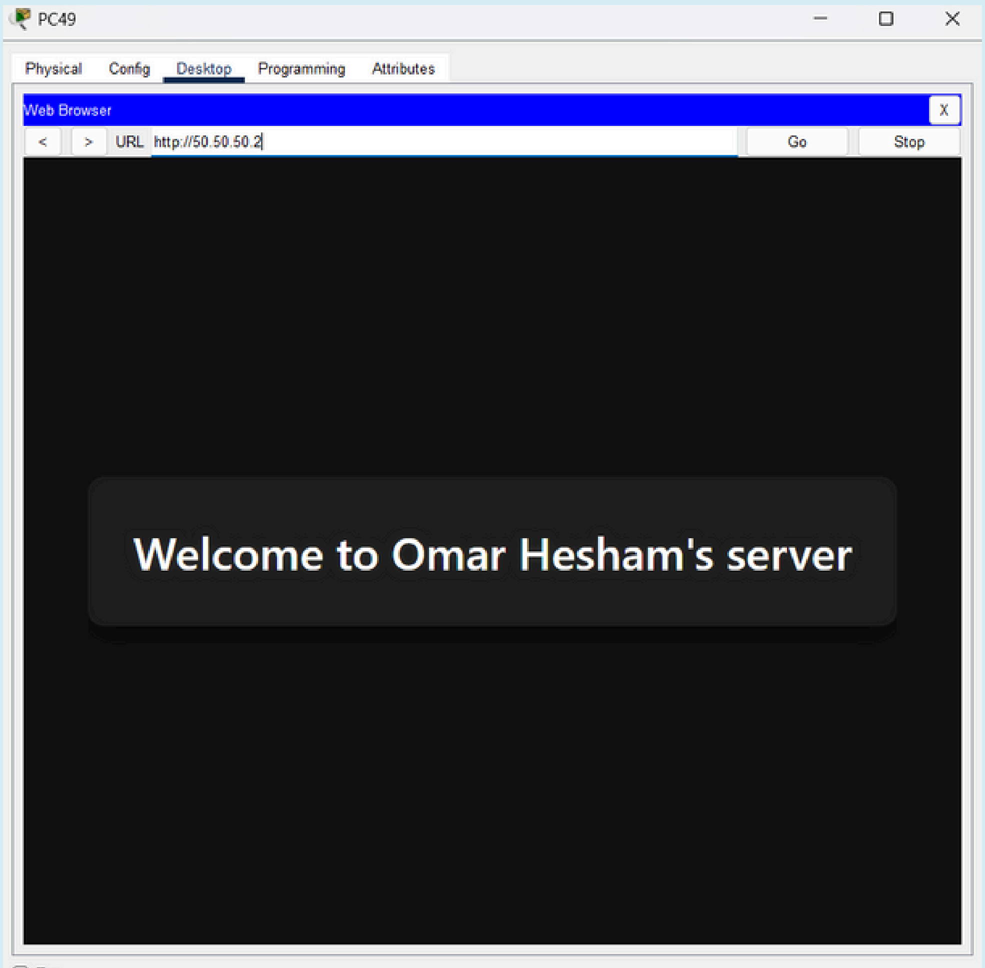


Configuration on Router 11

```
interface GigabitEthernet0/0/1
 ip address 10.10.10.1 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/1/0
 ip address 70.70.70.2 255.255.255.252
 ip nat outside
!
interface Serial0/1/1
 ip address 60.60.60.2 255.255.255.252
 ip nat outside
```

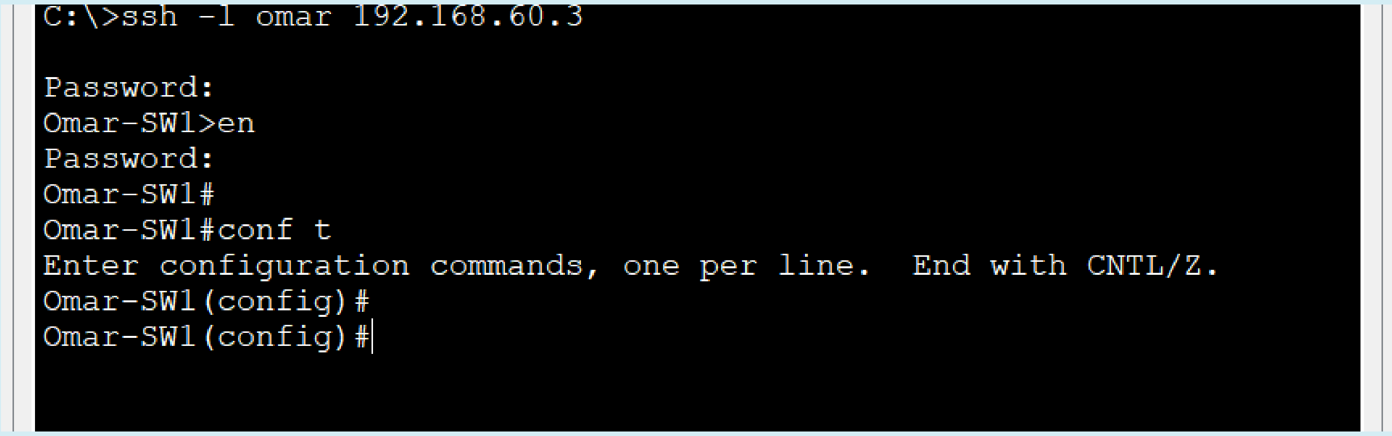
```
ip nat inside source static 10.10.10.2 50.50.50.2
ip classless
```

Can be accessed from any host at any branch

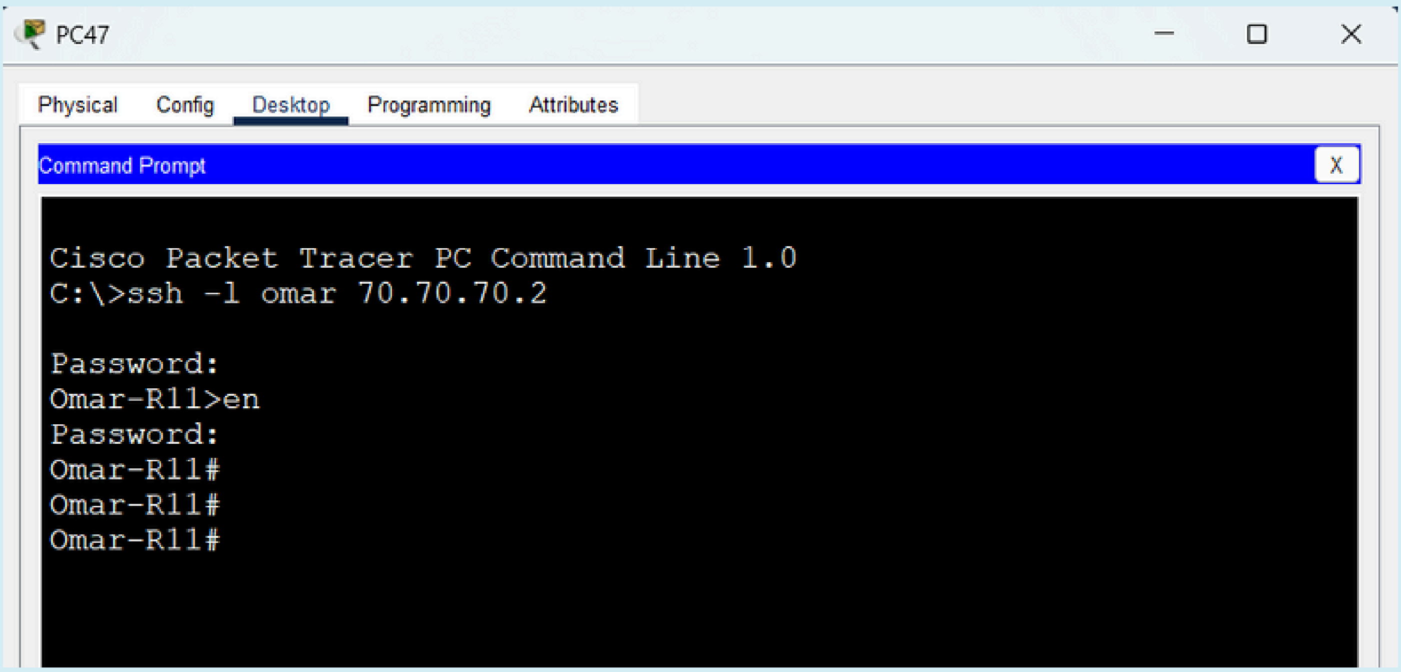


Secure Shell (SSH)

- Secure Remote Management (SSH)
- To ensure secure and confidential administrative access, Secure Shell (SSH) is enabled on all routers and switches throughout the network.
- This implementation replaces insecure, clear-text protocols like Telnet. By using SSH, all remote management sessions—including command entry and device output—are fully encrypted.
- Furthermore, access via SSH is integrated with the central AAA RADIUS server. This means that no administrator can open an SSH session without first being successfully authenticated by the RADIUS server, combining transport-level encryption (SSH) with centralized user control (AAA).



SSH to switch 1



SSH to router 11

Omar's Team

CCNA FINAL PROJECT

Supervised By

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